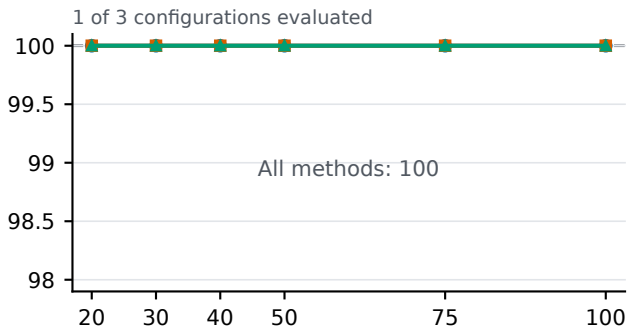


RULER 64K: retrieval subtypes

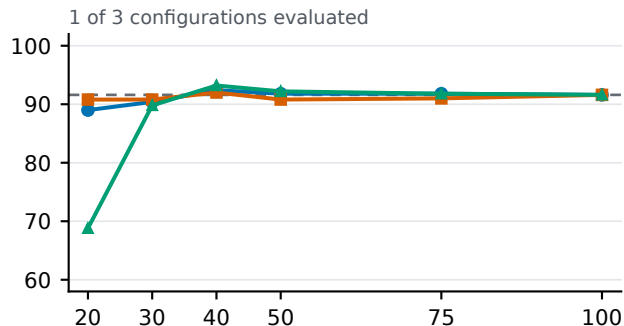
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

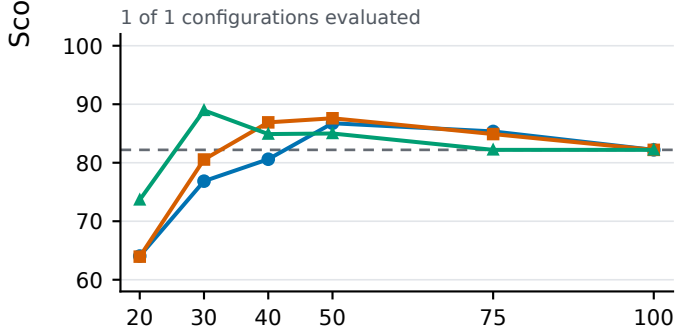
Single NIAH



Multi-keys NIAH



Multi-values NIAH



Retention ratio (%)

Categories follow RULER Section 3. Single and multi-key means combine their available variants. Multi-query retrieval is unevaluated at 64K and has no curve. Only the seven evaluated 64K tasks are included. A one-configuration family reproduces its individual task curve.